WASP-1b

R-1345

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-1_161205` · created 2026-07-31T22:01:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457727.714 +/- 0.0078 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.068 +/- 0.027
Transit depth (Rp/Rs)^2	0.47 +/- 0.37 [%]
Orbital Inclination (inc)	89.85 +/- 1.45
Airmass coefficient 1 (a1)	1.014 +/- 0.015
Airmass coefficient 2 (a2)	-0.0055 +/- 0.0024
Scatter in the residuals of the lightcurve fit is	1.45 %
Best Comparison Star	#3 - [181, 237]
Optimal Aperture	4.09
Optimal Annulus	9.54
Transit Duration (day)	0.1104 +/- 0.0084

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.068 ± 0.027 vs 0.1036 (deviation 0.83σ)
Duration fit vs published	0.1104 d vs 0.1567 d (deviation 3.48σ)
Mid-time O–C	0.0 min
Depth SNR	1.6
Residual scatter	1.45 %

Light curve

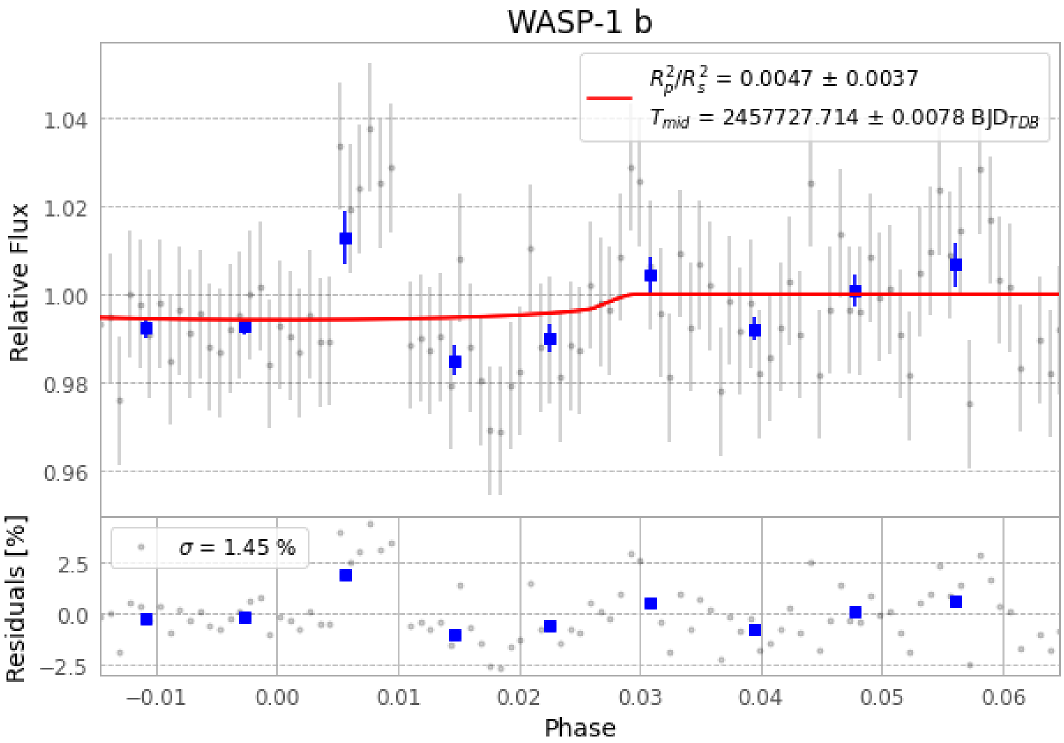

Figure 1 — FinalLightCurve_WASP-1 b_2016-12-04.png (SHA-256 9f3394fae7cfc3f1…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [175, 277], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 5956f30038a73602…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

97 FITS frames (64 MB), `WASP-1161205040912.FITS` → `WASP-1161205085712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-161205.log (a04fb926e25f…), FinalLightCurve_WASP-1 b_2016-12-04.png (9f3394fae7cf…), FinalLightCurve_WASP-1 b_2016-12-04.csv (b8524ab23443…)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)